İzmir Institute of Technology

Department of Computer Engineering

CENG424 – Embedded Computer Systems

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Homework 4

Servo Motor Position Control Using PID Controller

Name & Surname: Zekiye Ayperi Tatar

Student ID: 310206043

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İzmir Institute of Technology, Urla, İzmir, Turkey

1. Introduction

In this homework, a PID controller was used to control the angular position of a servo motor in Simulink. The aim was to move the motor from the initial position of 0 rad to the desired position of π rad and observe the response of the system.

2. System Model

The servo motor dynamics are given by

$$J\ddot{\theta}(t) + B\dot{\theta}(t) = K_t u(t) \tag{1}$$

where $\theta(t)$ is the angular position of the motor and $u(t)$ is the input control signal.

The parameter values used in the simulation are given in Table 1.

Table 1: Servo motor parameters

Parameter	**Symbol**	**Value**	**Unit**
Moment of inertia	J	0.005	kg·m^2
Damping coefficient	B	0.6	N·m·s/rad
Torque constant	K_t	0.08	N·m/A
Desired position	$\theta_{desired}$	π	rad
Initial position	$\theta(0)$	0	rad
K_t/J	K_{dc}	16	–
B/J	B_{dc}	120	–

The differential equation was rearranged as

$$\ddot{\theta}(t) = \frac{K_t}{J}u(t) - \frac{B}{J}\dot{\theta}(t) \tag{2}$$

Using the given values,

$$\ddot{\theta}(t) = 16u(t) - 120\dot{\theta}(t) \tag{3}$$

The transfer function of the servo motor is

$$\frac{\Theta(s)}{U(s)} = \frac{K_t}{Js^2 + Bs} = \frac{0.08}{0.005s^2 + 0.6s} = \frac{16}{s^2 + 120s} \tag{4}$$

3. PID Controller Design

A PID controller block was used in the closed-loop system. The error signal is

$$e(t) = \theta_{desired} - \theta(t) \tag{5}$$

and the PID controller generates the control signal $u(t)$ based on this error.

The controller parameters were tuned using the PID Tuner in Simulink. The main purpose of the tuning process was to obtain a fast response with small overshoot and almost zero steady-state error.

4. Simulink Model

The Simulink model was built as a closed-loop control system. A constant block was used for the reference input $\theta_{desired} = \pi$. The output $\theta(t)$ was fed back and subtracted from the reference to generate the error. Then, the PID controller output was passed through the gain block $K_t/J = 16$. After that, the damping term $(B/J)\dot{\theta}$ was subtracted, and two integrator blocks were used to obtain $\dot{\theta}(t)$ and $\theta(t)$.

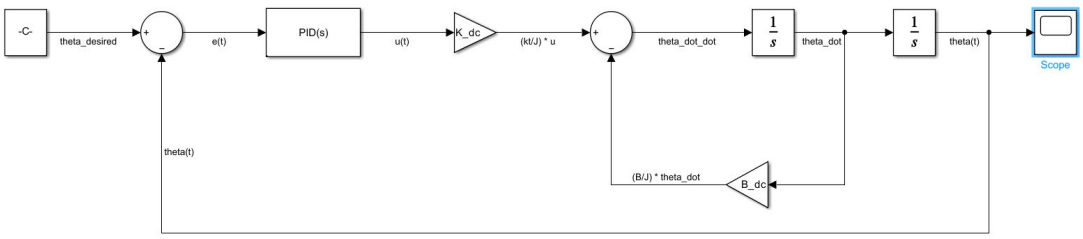

Figure 1: Simulink model of the servo motor position control system

5. Simulation Results

The simulation was run for 0.5 seconds, which was enough to observe both the transient response and the steady-state behavior.

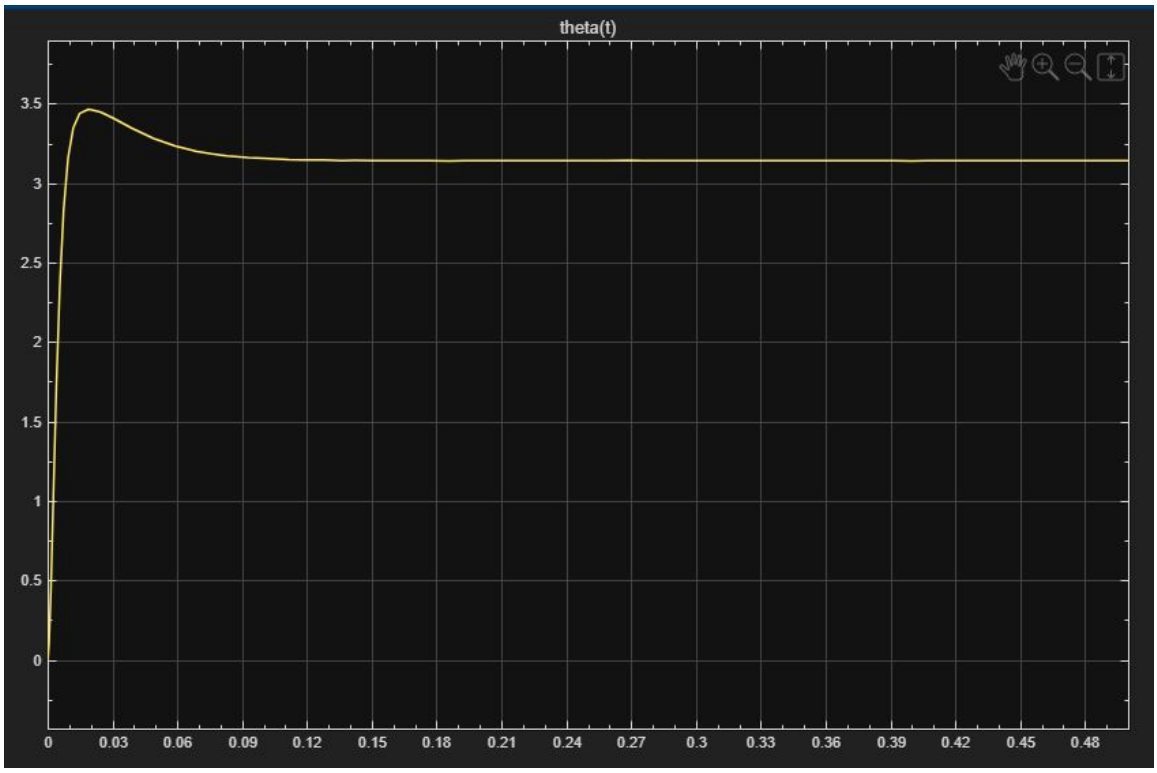

Figure 2: Scope output of $\theta(t)$

From the simulation result, the output reaches the desired position of π rad quickly. There is a small overshoot at the beginning, but the response settles after a short time. The final value stays very close to π, which means the steady-state error is approximately zero.

Table 2: Performance summary

Metric	**Approximate Value**	**Comment**
Rise time	0.01–0.02 s	The response reaches the target very quickly
Peak value	3.42 rad	Maximum value reached by the output
Overshoot	8.9%	Small overshoot before settling
Settling time	0.15–0.20 s	Output becomes stable in a short time
Steady-state value	$\approx \pi$ rad	Final value is close to the reference
Steady-state error	≈ 0 rad	No significant residual error

6. Performance Evaluation

The position accuracy of the system is good because the output settles at the desired value of π rad. The response time is also fast since the motor reaches the target in a short time. Even though there is some overshoot, it is not very large and the system becomes stable quickly. In general, the PID controller gives a satisfactory result for this servo motor position control problem.

7. Conclusion

In this homework, a PID-controlled servo motor system was modeled and simulated in Simulink. The results show that the controller can move the motor from 0 rad to π rad successfully. The response is fast, the overshoot is limited, and the steady-state error is almost zero.

Appendix: MATLAB Parameter Script

```
J = 0.005;
B = 0.6;
Kt = 0.08;

theta_desired = pi;
theta0 = 0;
theta_dot0 = 0;

K_dc = Kt / J;
B_dc = B / J;
```